



**Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember***

Laporan Akhir Praktikum Jaringan Komputer

Firewall & NAT

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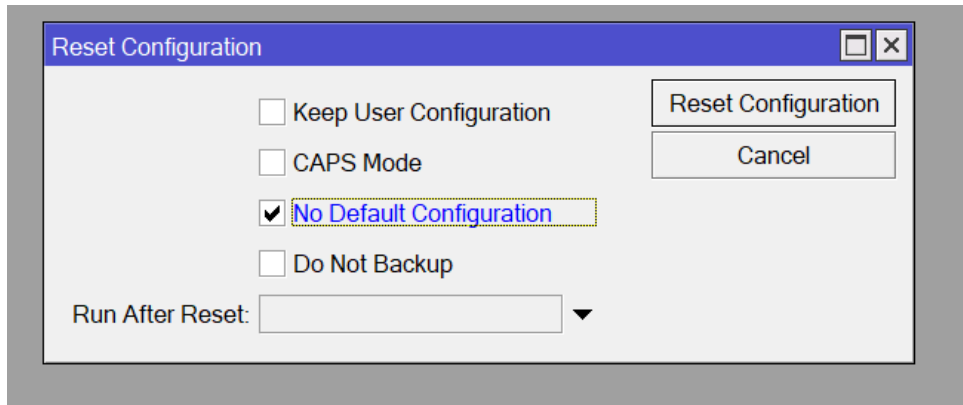
2026

1 Langkah-Langkah Percobaan

1.1 Praktikum 1

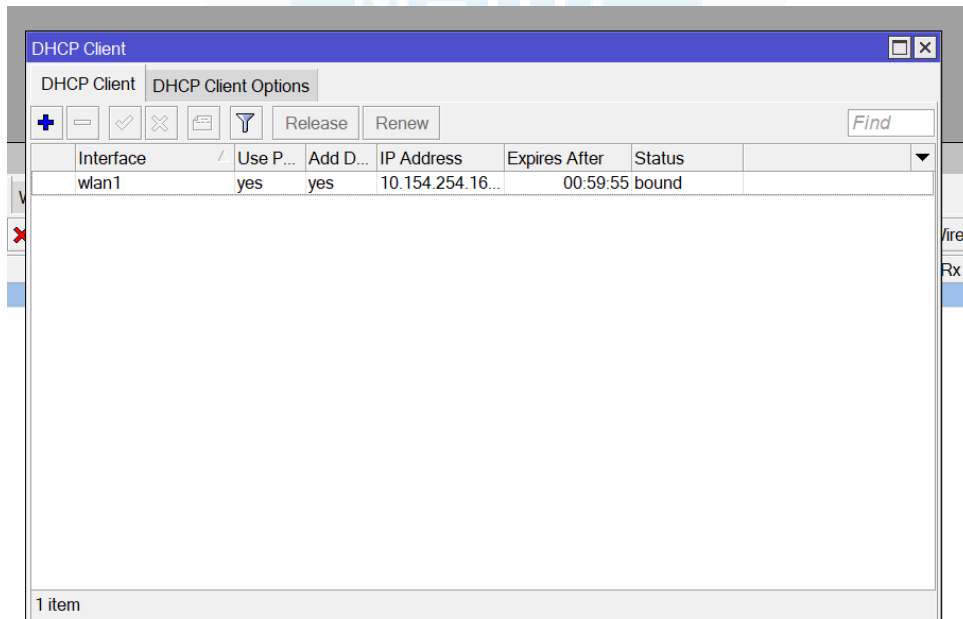
1.1.1 Reset Router

Buka aplikasi Winbox lalu hubungkan ke router, masuk menu System -> Reset COnfiguration, centang No Default Configuration lalu klik reset configuration



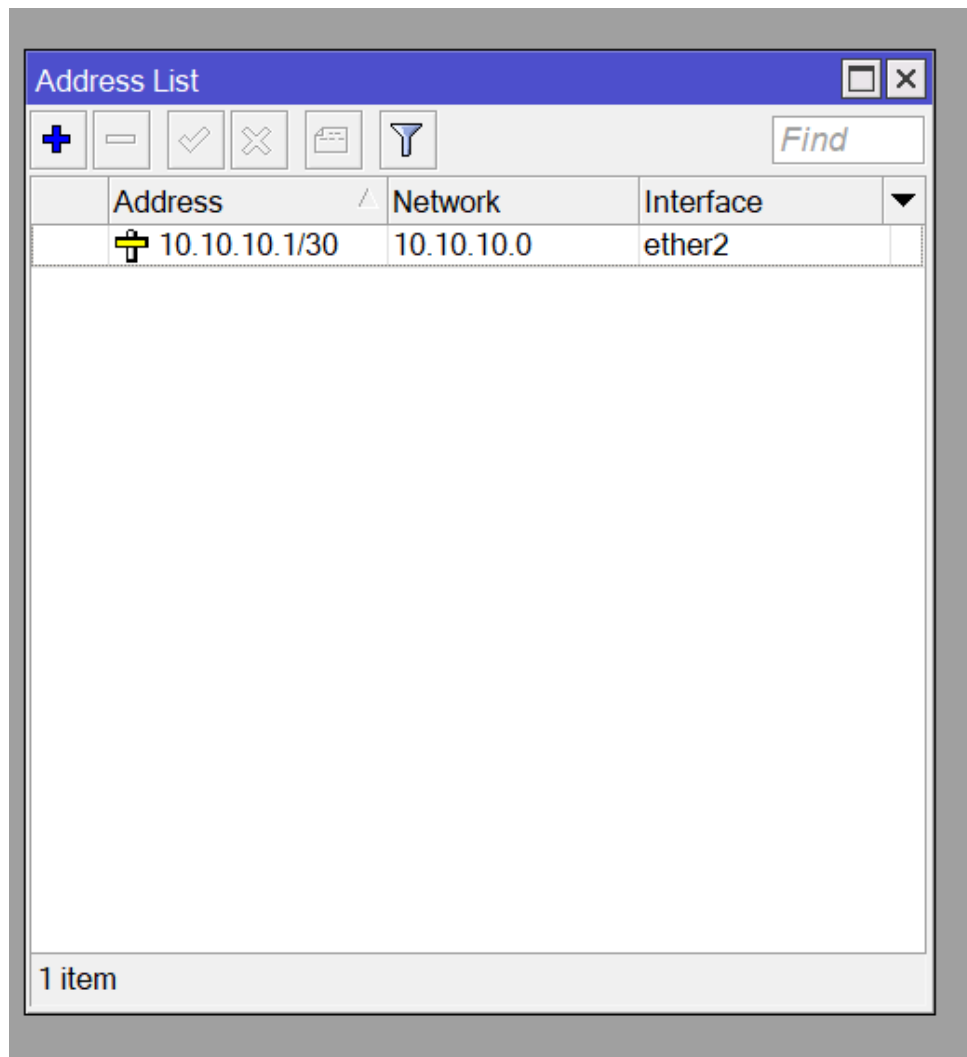
1.1.2 Konfigurasi DHCP client pada Router A

Masuk menu IP -> DHCP Client -> klik tombol + -> pilih interface ether1 (atau wlan1 karena saat praktikum internet ITS tidak bisa) -> klik apply dan OK -> pastikan status menjadi Bound.



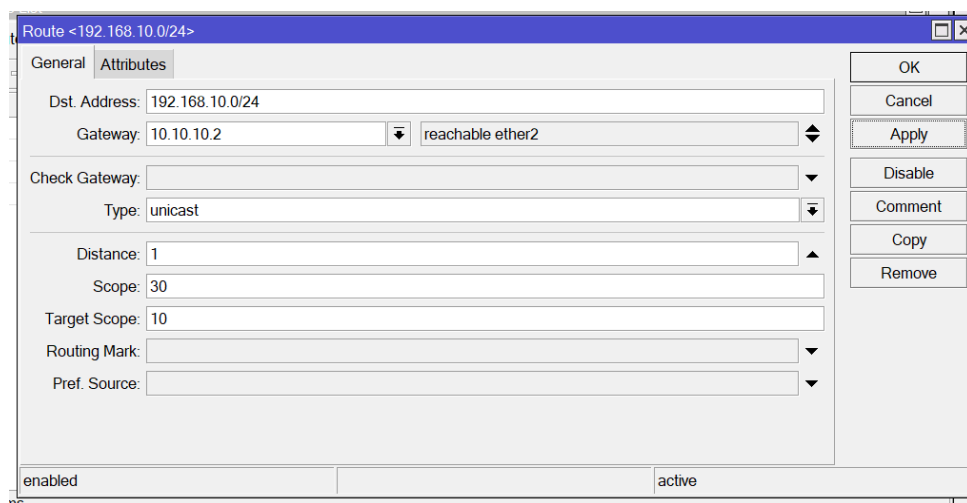
1.1.3 Konfigurasi IP Address Router A

Masuk Menu IP -> Addresses -> klik tombol + -> isi dengan Address: 10.10.10.1/30 dan interface: ether2 -> klik Apply dan OK.



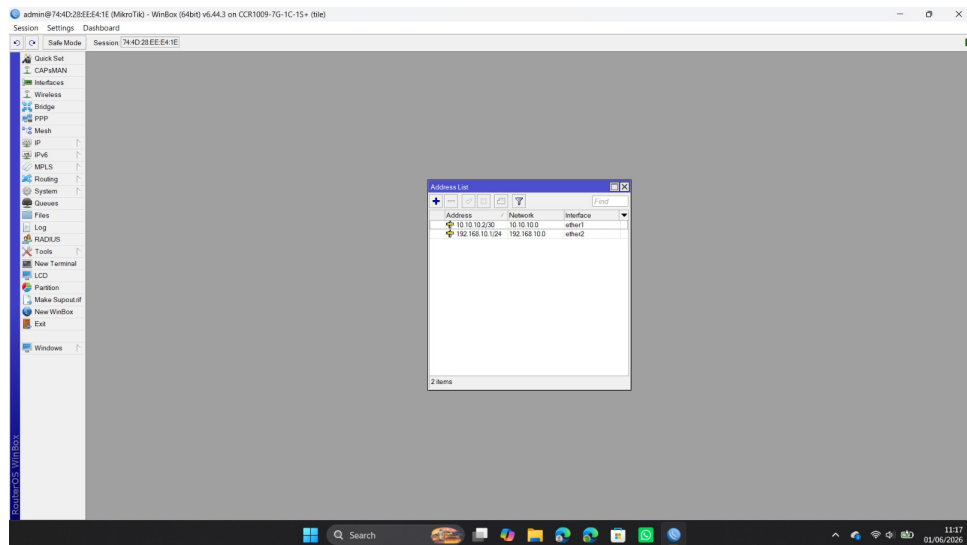
1.1.4 Konfigurasi Routing Statis Router A

Masuk menu IP -> Routes -> klik tombol + -> isi Dst.Address: 192.168.10.0/24 dan gateway: 10.10.10.2
-> Klik Apply dan OK.



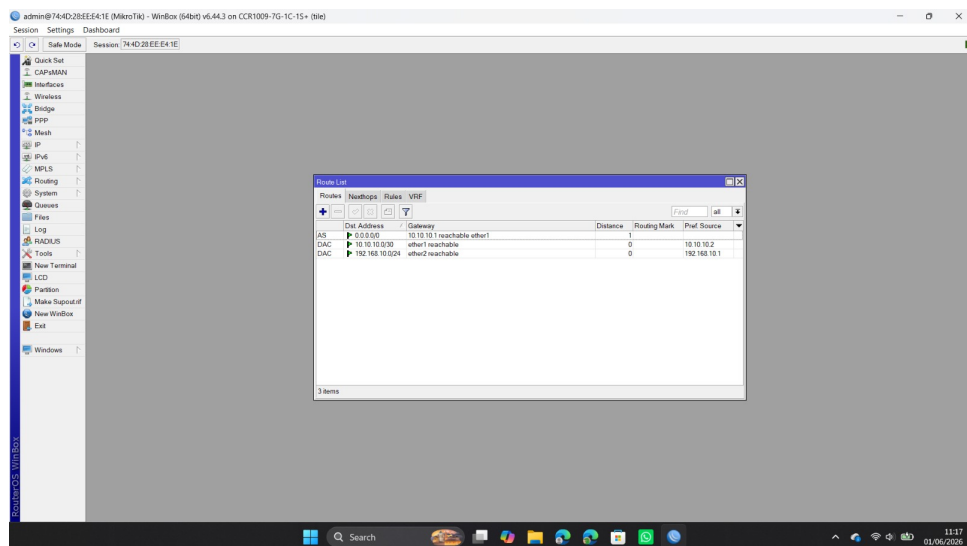
1.1.5 Konfigurasi IP address router B

Buka menu IP -> Addresses -> klik tombol + -> isi dengan Address: 192.168.10.1/24 dan Interface: ether2 -> klik Apply dan OK. Lalu tambahkan lagi dengan isi Address: 10.10.10.2/30 dan Interface: ether1.



1.1.6 Konfigurasi route Router B

Masuk menu IP -> Routes -> Klik tombol + -> Isi dengan Dst.Address: 0.0.0.0/0 dan Gateway: 10.10.10.1 -> Klik Apply dan OK.



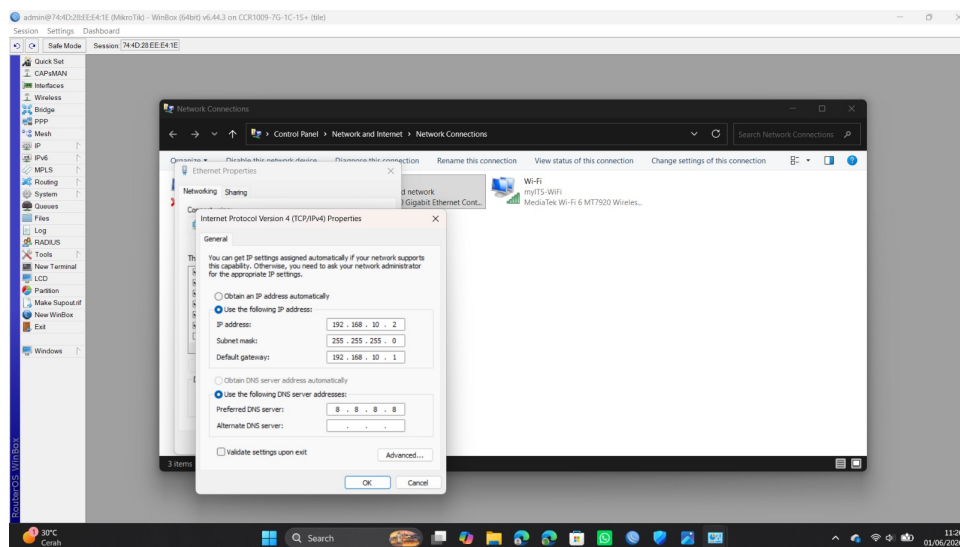
1.1.7 Konfigurasi NAT Masquerade Router A

Masuk menu IP -> Firewall -> tab NAT -> klik tombol + -> isi dengan Chain: srcnat dan Out Interface: ether1/wlan1 -> tab Action isi dengan Action: Masquerade -> Klik Apply dan OK.

Filter Rules NAT Mangle Raw Service Ports Connections Address Lists Layer7 Protocols											
+ [-] ✓ ✗ 📄 🔍 Reset Counters 00 Reset All Counters Find all											
#	Action	Chain	Src. Address	Dst. Address	Proto...	Src. Port	Dst. Port	In. Inter...	Out. Int...	Bytes	Packets
0	mas...	srcnat							wlan1	0 B	0

1.1.8 Konfigurasi IP statis pada Laptop

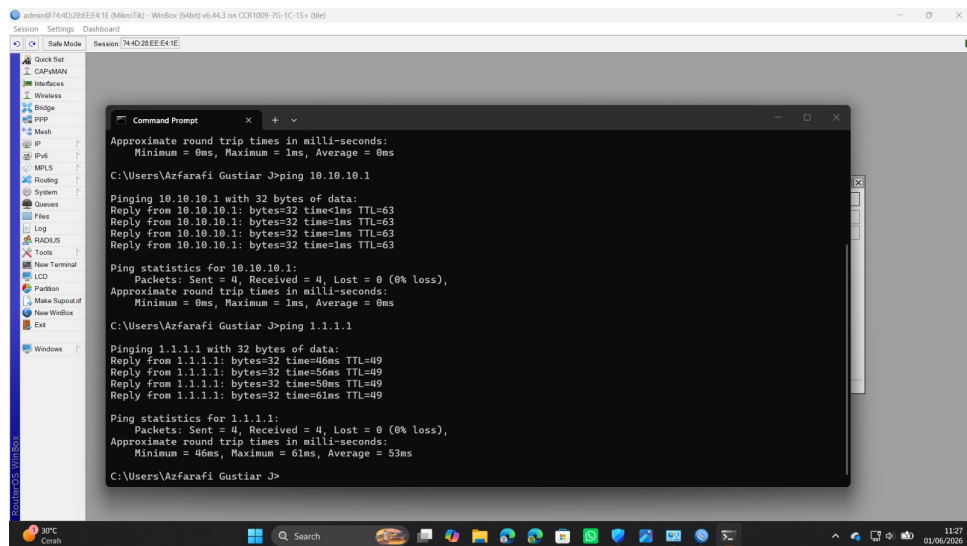
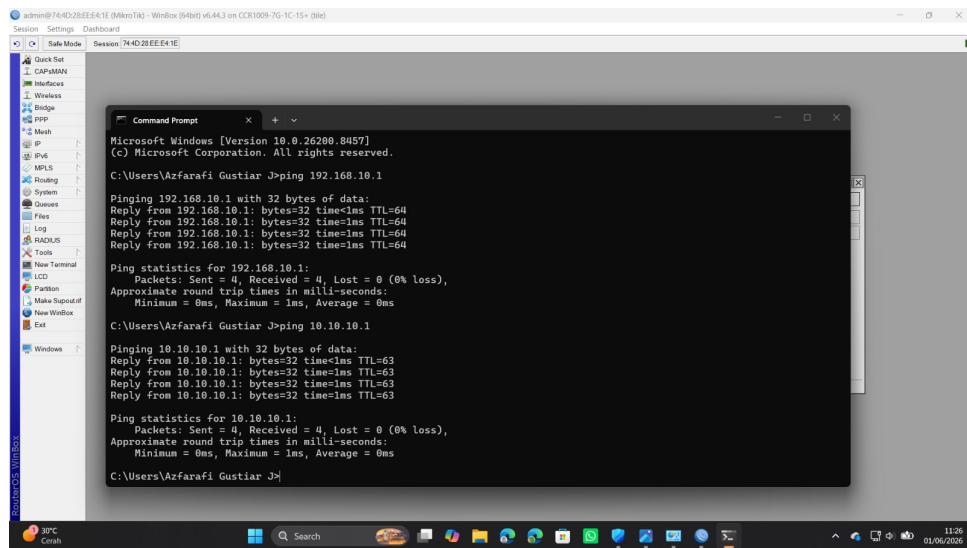
Buka Control Panel -> Network & Internet -> Network and Sharing Center -> Change Adapter Settings -> Buka setting Adapter Ethernet -> Properties -> IPV4 Properties -> isi dengan IP: 192.168.10.2 & Subnet Mask: 255.255.255.0 & Gateway: 192.168.10.1 & DNS: 8.8.8.8 -> Klik OK.



1.1.9 Pengujian Jaringan

Buka Command Prompt lalu lakukan

- ping 192.168.10.1 (Router B)
- ping 10.10.10.1 (Router A)
- ping 1.1.1.1 (Internet)

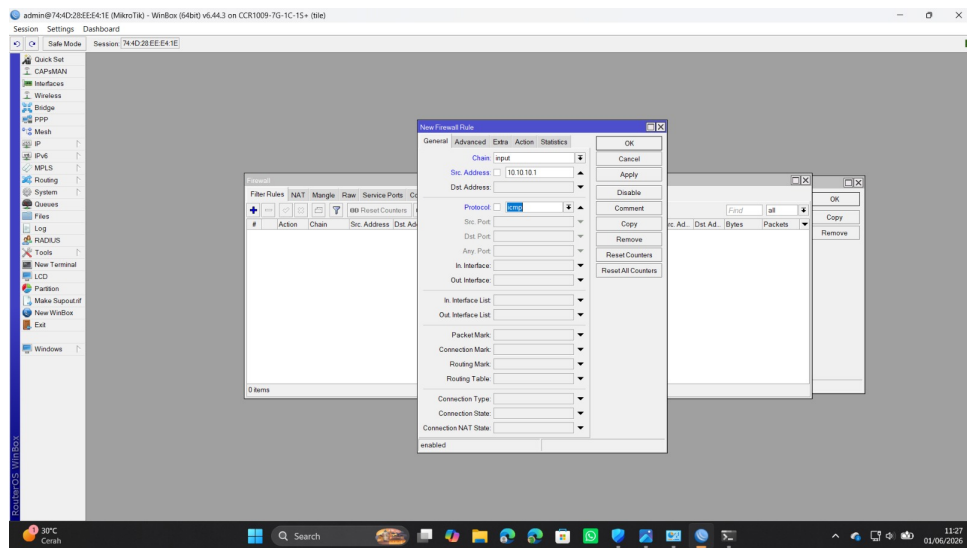


1.2 Praktikum 2

1.2.1 Konfigurasi Filter Firewall

1. Konfigurasi Filter chain Input

Masuk menu IP -> Firewall -> Filter Rules -> Klik Tombol + -> isi dengan Chain: Input & Src. Address: 10.10.10.1 & Protocol: icmp & Action: drop



```

Terminal <1>
[?] Gives the list of available commands
command [?] Gives help on the command and list of arguments
[Tab] Completes the command/word. If the input is ambiguous,
a second [Tab] gives possible options
/ Move up to base level
.. Move up one level
/command Use command at the base level
[admin@MikroTik] > ping 10.10.10.2
    SEQ HOST                SIZE TTL TIME  STATUS
    0 10.10.10.2              56 127 3ms  timeout
    1 10.10.10.2              56 127 3ms  timeout
    2 10.10.10.2              56 127 3ms  timeout
    sent=3 received=0 packet-loss=100%

[admin@MikroTik] > ping 192.168.10.2
    SEQ HOST                SIZE TTL TIME  STATUS
    0 192.168.10.2            56 127 3ms
    1 192.168.10.2            56 127 3ms
    2 192.168.10.2            56 127 3ms
    sent=3 received=3 packet-loss=0% min-rtt=3ms avg-rtt=3ms max-rtt=3ms

[admin@MikroTik] >

```

2. Konfigurasi Filter chain Forward

(Disable Filter Rule sebelumnya terlebih dahulu)

Masuk menu IP -> Firewall -> Filter Rules -> Klik Tombol + -> isi dengan Chain: Forward & Src.

Address: 10.10.10.1 & Protocol: icmp & Action: drop

```

Terminal <1>
MikroTik RouterOS 6.42.1 (c) 1999-2018      http://www.mikrotik.com/

[?]          Gives the list of available commands
command [?]  Gives help on the command and list of arguments

[Tab]        Completes the command/word. If the input is ambiguous,
              a second [Tab] gives possible options

/            Move up to base level
..          Move up one level
/command     Use command at the base level
[admin@MikroTik] > ping 10.10.10.2
      SEQ HOST                                SIZE TTL TIME  STATUS
      0 10.10.10.2                            56  64 0ms
      1 10.10.10.2                            56  64 0ms
      sent=2 received=2 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=0ms

[admin@MikroTik] > ping 192.168.10.2
      SEQ HOST                                SIZE TTL TIME  STATUS
      0 192.168.10.2                          timeout
      1 192.168.10.2                          timeout
      sent=2 received=0 packet-loss=100%

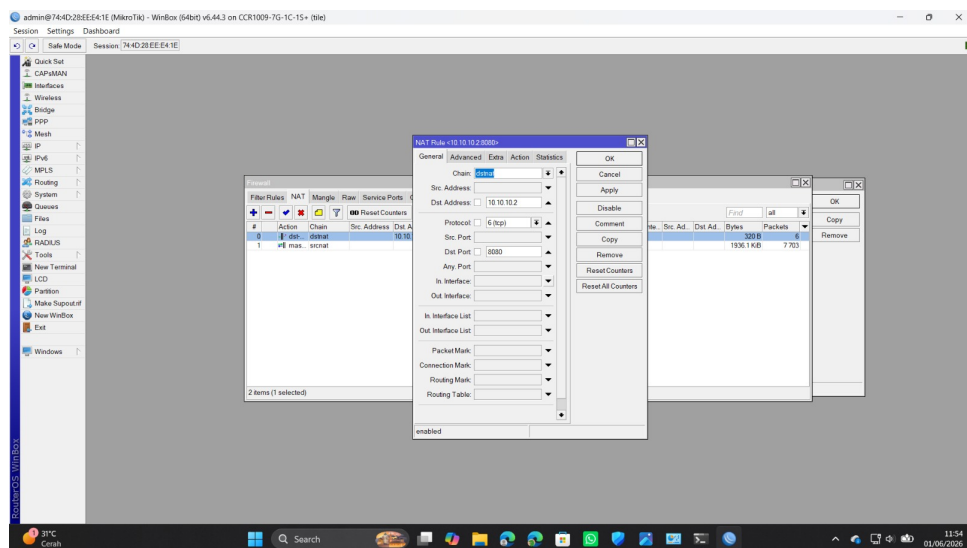
[admin@MikroTik] >

```

1.3 Praktikum 3

1.3.1 Konfigurasi NAT

Masuk menu IP -> Firewall -> tab NAT -> klik tombol + -> tab General isi dengan Chain: dstnat, Dst. Address: 10.10.10.2, Protocol: tcp, Dst. Port: 8080 -> Tab Action isi dengan Action: dst-nat, To Addresses: 192.168.10.2, To Ports: 80 -> Klik Apply dan OK.



1.3.2 Web Server pada Laptop

Jalankan perintah berikut pada Command Prompt:
python -m http.server 80

1.3.3 Pengujian NAT Port Forward dari Router A

Pada Terminal Router A Jalankan:

/tool fetch url="http://10.10.10.2:8080/" keep-result=no


```
Terminal <1>
MMMM MMMM KKK TTTTTTTTTT KKK
MMM MMMM MMM III KKK KKK RRRRRR OOOOOO TTT III KKK KKK
MMM MM MMM III KKKKK RRR RRR OOO OOO TTT III KKKKK
MMM MMM III KKK KKK RRRRRR OOO OOO TTT III KKK KKK
MMM MMM III KKK KKK RRR RRR OOOOOO TTT III KKK KKK

MikroTik RouterOS 6.42.1 (c) 1999-2018 http://www.mikrotik.com/

[?] Gives the list of available commands
command [?] Gives help on the command and list of arguments

[Tab] Completes the command/word. If the input is ambiguous,
a second [Tab] gives possible options

/ Move up to base level
.. Move up one level
/command Use command at the base level
[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080/" keep-result=no
status: finished
downloaded: 2KiB-C-z pause]
total: 2KiB
duration: 0s
[admin@MikroTik] >
```

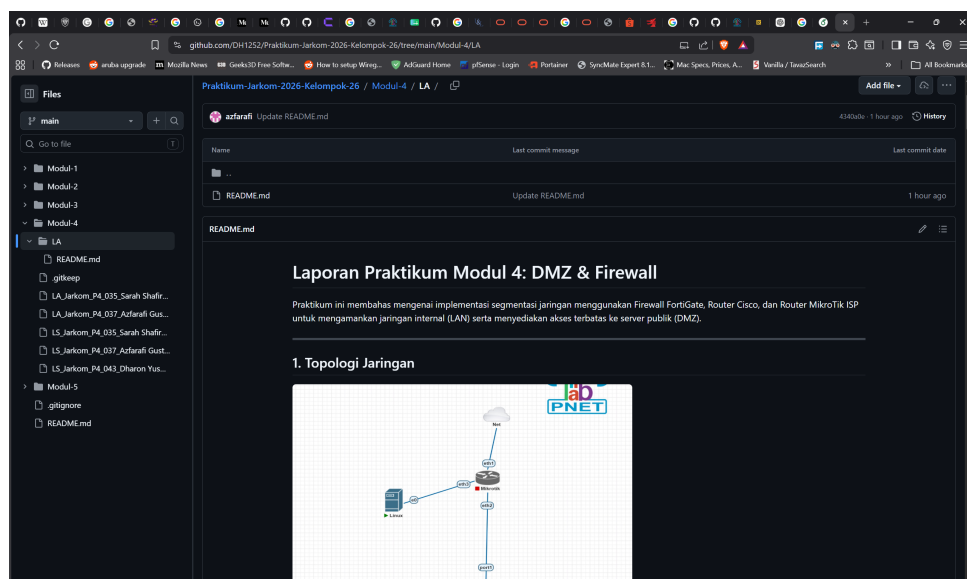
1.3.4 Pengecekan Koneksi

Buka menu IP -> Firewall -> Connections

2 Analisis Hasil Percobaan

Berdasarkan hasil praktikum yang didapatkan, percobaan konfigurasi NAT dan routing kedua router berhasil. Halangan yang ditemui adalah penggantian akses internet dari kabel ke WLAN bridge dikarenakan gangguan Jaringan Lab saat praktikum. Pengujian konektivitas antara Router A, Router B, Laptop dan Internet telah berhasil. Juga pengujian Firewall Filter dengan chain Input (Filter Router itu sendiri) dan Chain Forward (Filter paket yang melewati Router) juga telah dibuktikan dengan kegagalan ping dengan perangkat yang diblokir. Port Forwarding juga berhasil dengan mengalihkan port 8080 pada Router B ke port 80 pada IP laptop yang menjalankan Web Server.

3 Hasil Tugas Modul



4 Kesimpulan

Berdasarkan Praktikum yang dilakukan, Praktikum Konfigurasi routing, Network Address Translation (NAT), Firewall, dan port forwarding pada platform MikroTik telah berhasil dan sesuai dengan teori jaringan komputer. Hasil praktikum menunjukkan keberhasilan komunikasi antar jaringan, Penyaringan paket dengan Firewall Filter, serta Pembukaan service internal ke jaringan luar dengan Port Forwarding NAT

5 Lampiran

5.1 Dokumentasi saat praktikum

